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Assignment: Day 16 Practice Question
Subject: MongoDB Database Design

1. Product Catalog

MongoDB Collection: `products`

The product catalog stores information about all products available in the online retail system.

Schema Example

```
{
  "_id": "ObjectId",
  "name": "string",
  "category": "string",
  "price": "number",
  "stock": "number",
  "brand": "string",
  "attributes": {
    "color": "string",
    "size": "string",
    "material": "string"
  },
  "created_at": "date"
}
```

Sample Product Documents

```
{
  "name": "Wireless Headphones",
  "category": "Electronics",
  "price": 2999,
  "stock": 100,
  "brand": "Sony",
  "attributes": {
    "color": "Black",
    "battery_life": "30 hours"
  }
}
{
  "name": "Men's T-Shirt",
  "category": "Clothing",
  "price": 799,
  "stock": 200,
  "brand": "Nike",
  "attributes": {
    "size": "L",
    "color": "Blue",
    "material": "Cotton"
  }
}
{
  "name": "Atomic Habits",
  "category": "Books",
  "price": 499,
```

```
  "stock": 80,
  "author": "James Clear",
  "attributes": {
    "pages": 320,
    "language": "English"
  }
}
```

2. Customer Orders

MongoDB Collection: `orders`

This collection stores customer order details and purchased products.

Schema Example

```
{
  "_id": "ObjectId",
  "user_id": "ObjectId",
  "order_date": "date",
  "products": [
    {
      "product_id": "ObjectId",
      "name": "string",
      "quantity": "number",
      "price": "number"
    }
  ],
  "total_amount": "number",
  "payment_status": "string"
}
```

Sample Order Document

```
{
  "user_id": "U1001",
  "order_date": "2026-03-04",
  "products": [
    {
      "product_id": "P101",
      "name": "Wireless Headphones",
      "quantity": 1,
      "price": 2999
    },
    {
      "product_id": "P102",
      "name": "Men's T-Shirt",
      "quantity": 2,
      "price": 799
    }
  ],
  "total_amount": 4597,
  "payment_status": "Paid"
}
```

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3. User Authentication

MongoDB Collection: `users`

This collection stores user account details for authentication.

Schema Example

```
{
  "_id": "ObjectId",
  "username": "string",
  "email": "string",
  "password_hash": "string",
  "created_at": "date",
  "role": "string"
}
```

Sample User Document

```
{
  "username": "naitik123",
  "email": "naitik@gmail.com",
  "password_hash": "hashed_password_123",
  "created_at": "2026-03-04",
  "role": "customer"
}
```

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4. Querying and Indexing

Indexes improve database query performance.

Create Index for Product Category

```
db.products.createIndex({ category: 1 })
```

Query Products by Category

```
db.products.find({ category: "Electronics" })
```

Create Index for User Orders

```
db.orders.createIndex({ user_id: 1 })
```

Query Orders for Specific User

```
db.orders.find({ user_id: "U1001" })
```

Create Index for Product Price

```
db.products.createIndex({ price: 1 })
```

Query Products Sorted by Price

```
db.products.find().sort({ price: 1 })
```